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Journal of Synthetic Astrophysics, 12, 101–118 (2023)

An I-band Period–Luminosity Calibration for Galactic Field RR Lyrae Stars from Trigonometric Parallaxes

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Abstract. We present a new calibration of the I-band period–luminosity (PL) relation for RR Lyrae stars based on parallaxes for 214 nearby field stars. Fitting fundamental-mode and fundamentalized first-overtone pulsators jointly, we obtain $M_I = (-1.13 \pm 0.06)\log P_F + (0.47 \pm 0.04)$ at fixed metallicity, with an intrinsic scatter of 0.04 mag. The I-band relation is substantially less sensitive to reddening and evolutionary effects than the V-band luminosity–metallicity relation, making it well suited for distances to old stellar systems.

1. Introduction

RR Lyrae stars are core-helium-burning, horizontal-branch pulsators found in old stellar populations. In the optical V band their absolute magnitudes are nearly independent of period, but toward redder bands a clear period–luminosity relation emerges, driven by the bolometric correction's dependence on effective temperature. This work provides an updated I-band zero point.

2. Sample and Method

We selected stars with parallax uncertainties below 8% and well-sampled light curves. First-overtone (RRc) periods were fundamentalized using $\log P_F = \log P_{1O} + 0.127$. A Bayesian hierarchical fit accounts for parallax zero-point offsets and Lutz–Kelker-type biases.

3. Results

The adopted relation is $M_I = -1.13 \log P_F + 0.47$. Including a metallicity term yields a coefficient of 0.18 ± 0.05 mag dex^{-1}, which we recommend when [Fe/H] is known. Systematic uncertainty in the zero point is estimated at 0.05 mag.

Table 1. Summary of key values.

Coefficient	Value	Uncertainty
Slope a	−1.13	0.06
Zero point b	0.47	0.04
Scatter σ	0.04 mag	—
N stars	214	—

References

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